

# EE 210L: Digital Circuits and Logic Design Lab

## Lab 2: Boolean Simplification

Name: \_\_\_\_\_ Date: \_\_\_\_\_

Partner(s): \_\_\_\_\_ Section: \_\_\_\_\_

*Record logic levels only, 0 or 1. Enter your pre-lab predictions before you measure, and leave them as written even if a measurement disagrees.*

### Part 5.1: Power Supply Setup

Measured power supply voltage: \_\_\_\_\_ V

Current limit setting: \_\_\_\_\_ mA

### Part 5.2: Original Circuit (a)

| # | A | B | C | D | E | Predicted F | Measured F |
|---|---|---|---|---|---|-------------|------------|
| 1 | 0 | 0 | 0 | 0 | 0 |             |            |
| 2 | 1 | 0 | 0 | 0 | 0 |             |            |
| 3 | 0 | 0 | 0 | 0 | 1 |             |            |
| 4 | 0 | 1 | 1 | 1 | 0 |             |            |
| 5 | 0 | 1 | 1 | 0 | 0 |             |            |

Did every row match your prediction? If not, what was wrong? \_\_\_\_\_

### Part 5.3: Simplified Circuit (a)

Simplified expression  $F =$  \_\_\_\_\_

*Draw your simplified circuit in the box. Label every chip and pin number you used.*

Figure 1: Simplified form of circuit (a).

| # | A | B | C | D | E | Measured F | Same as 5.2? |
|---|---|---|---|---|---|------------|--------------|
| 1 | 0 | 0 | 0 | 0 | 0 |            |              |
| 2 | 1 | 0 | 0 | 0 | 0 |            |              |
| 3 | 0 | 0 | 0 | 0 | 1 |            |              |
| 4 | 0 | 1 | 1 | 1 | 0 |            |              |
| 5 | 0 | 1 | 1 | 0 | 0 |            |              |

#### Part 5.4: Original Circuit (b)

| # | A | B | Predicted Y | Measured Y |
|---|---|---|-------------|------------|
| 1 | 0 | 0 |             |            |
| 2 | 0 | 1 |             |            |
| 3 | 1 | 0 |             |            |
| 4 | 1 | 1 |             |            |

Did every row match your prediction? If not, what was wrong? \_\_\_\_\_

#### Part 5.5: Simplified Circuit (b)

Simplified expression  $Y =$  \_\_\_\_\_

*Draw your simplified circuit in the box. Label every chip and pin number you used.*

Figure 2: Simplified form of circuit (b).

| # | A | B | Measured Y | Same as 5.4? |
|---|---|---|------------|--------------|
| 1 | 0 | 0 |            |              |
| 2 | 0 | 1 |            |              |
| 3 | 1 | 0 |            |              |
| 4 | 1 | 1 |            |              |

### Part 5.6: Gate and Chip Count

| Circuit        | AND gates | OR gates | NOT gates | Total gates | Packages |
|----------------|-----------|----------|-----------|-------------|----------|
| (a) original   |           |          |           |             |          |
| (a) simplified |           |          |           |             |          |
| (b) original   |           |          |           |             |          |
| (b) simplified |           |          |           |             |          |

How many gates does the simplified circuit (b) need, against the seven in the original? What does that tell you about reading a schematic at face value?

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### Discussion and Conclusions

Summarize what you measured and what it confirmed. Address at least: whether the simplified circuits reproduced the originals on every row you tested, which law did the most work in each simplification, and why testing five of the thirty-two possible input combinations of circuit (a) is not the same as proving the two circuits are equal.

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